

A Technical Introduction to Prompt engineering

Section 1

MIPRO (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. GEPA (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

Section 2

Prompt injection is a cybersecurity exploit in which adversaries craft inputs that appear legitimate but are designed to cause unintended behavior in machine learning models, particularly large language models. This attack takes advantage of the model's inability to distinguish between developer-defined prompts and user inputs, allowing adversaries to bypass safeguards and influence model behaviour. While LLMs are designed to follow trusted instructions, they can be manipulated into carrying out unintended responses through carefully crafted inputs.

Section 3

Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

Section 4

A text-to-image prompt commonly includes a description of the subject of the art, the desired medium (such as digital painting or photography), style (such as hyperrealistic or pop-art), lighting (such as rim lighting or crepuscular rays), color, and texture. Word order also affects the output of a text-to-image prompt. Words closer to the start of a prompt may be emphasized more heavily.

Section 5

Multi-shot A prompt may include a few examples for a model to learn from in context, an approach called few-shot learning. For example, the prompt may ask the model to complete "maison → house, chat → cat, chien →", with the expected response being dog.

Section 6

GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights, and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

Section 7

History Early precedents of structured user interaction with ruled-based AI systems can be found in enterprise automation software from 1990s. For example, The Intelligent Filing Manager (1999), developed by Krishna C. Mukherjee, used a dynamic Q&A interface driven by rule-based expert system to collect user inputs for generating regulatory filings automatically across jurisdictions. While not involving neural networks, such systems featured prompt-like workflows that influenced later human-in-the-loop AI designs. In 2018, researchers first proposed that all previously separate tasks in natural language processing (NLP) could be cast as question-answer problems over a context. In addition, they trained a first single, joint, multi-task model that would answer any task-related question like "What is the sentiment" or "Translate this sentence to German" or "Who is the president?" The AI boom saw an increased focus within academic literature and professional practice on applying prompting techniques to get the model to output the desired outcome and avoid nonsensical output, a process characterized by trial-and-error. After the release of ChatGPT in 2022, prompt engineering was soon seen as an important business skill; companies began hiring dedicated prompt engineers, although, given advances in AI's ability to generate prompts better than humans, the employment market for prompt engineers has faced uncertainty. According to The Wall Street Journal in 2025, the job of prompt engineer was one of the hottest in 2023, but has become obsolete due to models that better intuit user intent and to company trainings. A repository for prompts reported that over 2,000 public prompts for around 170 datasets were available in February 2022. In 2022, the chain-of-thought prompting technique was proposed by Google researchers. In 2023, several text-to-text and text-to-image prompt databases were made publicly available. The Personalized Image-Prompt (PIP) dataset, a generated image-text dataset that has been categorized by 3,115 users, has also been made available publicly in 2024.

Section 8

Automated prompt generation Recent research has explored automated prompt engineering, using optimization algorithms to generate or refine prompts without human intervention. These automated approaches aim to identify effective prompt patterns by analyzing model gradients, reinforcement feedback, or evolutionary processes, reducing the need for manual experimentation.

Section 9

Prompt A prompt is some natural language text that describes and prescribes the task that an artificial intelligence (AI) should perform. A prompt for a text-to-text language model can be a query, a command, or a longer statement referencing context, instructions, and conversation history. The process of prompt engineering may involve designing clear queries, refining wording, providing relevant context, specifying the style of output, and assigning a character for the AI to mimic in order to guide the model toward more accurate, useful, and consistent responses. When communicating with a text-to-image or a text-to-audio model, a typical prompt contains a description of a desired output such as "a high-quality photo of an astronaut riding a horse" or "Lo-fi slow BPM electro chill with organic samples". Prompt engineering may be applied to text-to-image models to achieve a desired subject, style, layout, lighting, and aesthetic.

Section 10

Using gradient descent to search for prompts In "prefix-tuning", "prompt tuning", or "soft prompting", floating-point vectors are searched directly by gradient descent to maximize the log-likelihood on outputs. An earlier result uses the same idea of gradient descent search, but is designed for masked language models like BERT, and searches only over token sequences, rather than numerical vectors. Formally, it searches for $\arg \max_{\tilde{X}} \sum_i \log \Pr[Y_i | X \sim * \tilde{X}_i]$ where $\tilde{X} \sim$ ranges over token sequences of a specified length.